AnalystLab Africa Experience Lab | FinTrust Digital Bank

Week 3 Documentation — Data Analytics Track

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PART A.

Review of Week 2 — Issues and Required Improvements

Issue/Gap	Evidence	Required Improvement
SQL median was only approximate.	MySQL has no MEDIAN() function; Week 2 used PERCENT_RANK as an approximation, cross-checked against Python rather than computed precisely in SQL.	Build an exact SQL median using ROW_NUMBER/window functions and quantify the approximation error.
"No relationship" findings not tested from a second angle	SQL Q7/Q8 (Week 2) found engagement score and preferred channel both flat against one comparison each	Test the same variables against different outcomes (e.g., engagement vs. channel mix, tenure vs. frequency) before concluding there is truly no pattern.
Time analysis stopped at the month level.	Week 2's day/hour charts showed uniform distribution; no second dimension (e.g., segment) was added.	Re-test day-of-week patterns within each customer segment.
The dashboard's risk section was one-factor-at-a-time.	Week 2 showed risk-flag rate by channel and by international status separately, never combined.	Build a combined view (international x value tier) to check for a compounding effect.
No recommendations existed yet	Week 2 produced findings (finding to evidence to business meaning) but stopped short of recommended actions.	Addressed directly in Part F below

PART B–C.

Advanced SQL and Python Analysis — Key Results

8 advanced SQL analyses and 5 advanced Python visualizations were completed (full queries, code, and write-ups in the accompanying .sql file and .ipynb notebook). The two most significant new results:

- **Value concentration:** 14.4% of transactions account for 67.4% of total value, and this top tier is flagged for risk review nearly twice as often as normal transactions (30.0% vs 17.9%). Visualized as a cumulative-value curve.
- **Compounding risk:** Transactions that are both international and high-value (≥NGN 100,000) are flagged at 50.8%, far above either factor alone (international ~36.9%, high-value 30.0%) or the baseline (17.1%). Visualized as a heatmap.

Three independent tests (engagement vs. frequency [Week 2], engagement vs. tenure, and engagement vs. channel mix) all confirmed no relationship between Digital_Engagement_Score and customer behavior, strengthening rather than weakening the original Week 2 observation. A new, mildly counterintuitive result also emerged: Failed transactions skew toward smaller amounts (mean NGN 39,406) than Successful, Pending, or Reversed (NGN 47,000–49,000), the opposite of a natural assumption that larger transactions fail more.

PART D.

Power BI Dashboard — Development from Week 2

Addition	Purpose
Matrix: Risk Flag Rate by International x Value Tier (with conditional-formatting heatmap)	Closes Gap #4: shows the compounding risk effect that single-factor visuals could not
High Value Share of Total Value (KPI card)	Surfaces the value-concentration finding (67.43%) as a headline metric.
Customer Transaction Frequency distribution (bar chart)	Shows the spread of customer activity, in contrast to the sharp concentration seen in transaction value
Month trend chart, rebuilt on its own page	Avoids cross-filtering the KPI cards when drilling into a month; axis range corrected to start at 0
The dashboard is restructured into 2 pages (Overview; Risk/Customer/Trend).	Keeps each page focused and readable rather than overcrowding a single page

PART E.

Validated Findings (summary)

Week 2 Finding	Verdict	What changed this week?
International status is the strongest risk signal.	Supported, deepened	Shown to compound with transaction value (50.8% when both present)
Deposit: low volume, high value	Supported	Shown to reflect a dataset-wide concentration pattern (Query 5), not a one-off
The "premium" segment does not mean the highest value.	Supported	Approximate median replaced with exact figures (within NGN 15–78 of the estimate)
Engagement score predicts nothing tested.	Strongly supported	1 test became 4 independent tests, all flat
Preferred channel does not predict actual use.	Unchanged	No new evidence gathered this week; carried forward as-is

PART F.

Management Recommendations (summary)

- **Flag "international + high-value" as a named, highest-priority risk category**, distinct from either factor alone.
- **Adopt value-weighted monitoring** rather than count-based monitoring, since a small share of transactions carries most financial exposure.
- **Clarify what the "Premium" segment is meant to represent**, since it does not correspond to the highest transaction value.
- **Review the definition of Digital_Engagement_Score** with its owning team; it does not predict transaction frequency or channel choice.
- **Use actual channel usage, not stated preference**, for channel investment and capacity decisions.

Testing / Validation Evidence Trail

Test/Validation	Finding	Action	Result	Note
Compared the exact SQL median to Week 2's PERCENT_RANK approximation.	Approximation was within NGN 15–78 per segment (under 1%).	Adopted exact ROW_NUMBER-based median going forward	Segment ranking unchanged; figures now precise	Closes Gap #1
Re-tested engagement score against tenure and channel mix	Both flat, no relationship found	Added a full-resolution scatter plot (Python) as a third, independent check	4 total independent tests agree: no relationship	Closes Gap #2
Re-tested day-of-week pattern within each customer segment	All segments stay within a narrow band; no weekday/weekend split	No cleaning action needed; finding confirmed at finer granularity	Week 2 "no time pattern" conclusion holds.	Closes Gap #3
Built a combined international x-value-tier risk view	Risk factors compound (50.8% vs 17.1% baseline).	Added a dedicated Matrix/heatmap visual to the dashboard	The dashboard now shows a risk pattern invisible in single-factor views.	Closes Gap #4
Validated matrix numbers against known totals	An active Channel=ATM slicer was silently filtering the whole page, producing a wrong grand total (21.18% instead of 19.60%).	Cleared the slicer and re-checked with an unfiltered card	Grand total confirmed at 19.60%, matching all prior checks	Reinforces checking for active filters before treating a number as wrong

Limitations and Assumptions

- Data remains synthetic; patterns may not reflect real banking behaviour.
- Only 3 months of data; no seasonality or long-term trend can be assessed.
- SQL's exact median (ROW_NUMBER) is now precise, but earlier-week figures quoting the PERCENT_RANK approximation differ slightly (documented above).
- Risk_Review_Flag remains a synthetic educational target, not a real fraud determination.
- Finding 5 (preferred channel) was not re-tested this week and is carried forward on Week 2 evidence only.